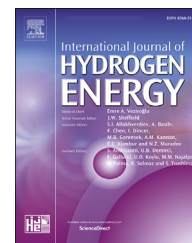


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Liquid organic hydrogen carriers (LOHCs) in the thermochemical waste heat recuperation systems: The energy and mass balances

Dmitry Pashchenko

Samara State Technical University, Molodogvardeiskaya 244 St, Samara 443100, Russia

HIGHLIGHTS

- Concept of thermochemical recuperation based on liquid organic hydrogen carriers (LOHC).
- Effect of pressure and temperature on energy and mass balance is studied.
- Temperature of waste heat for efficient using of LOHC is 300–350 °C.
- Effect of pressure on energy balance of the thermochemical recuperation system is negligible.

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ABSTRACT

This paper is presented a concept of thermochemical recuperation of waste heat based on hydrogen extraction from liquid organic hydrogen carriers (LOHC), on the example of methylcyclohexane-toluene system. The advantages of this concept is described, for example, a possibility to use a moderate low temperature of waste heat for generation high-exergy “green” hydrogen fuel. To understand the effect of operating parameters on the energy and mass balance, the thermodynamic analysis was performed. The chemical system for hydrogen generation was analyzed via Gibbs free energy minimization method. The thermodynamic analysis was conducted under various operating conditions: temperature of 100–400 °C, pressure of 1–4 bar. Aspen HYSYS software was used for the energy and mass conservation analysis. Sankey diagram for the energy flows is depicted. The results showed that the maximum energy efficiency the thermochemical waste heat recuperation system have in the temperature range above 300–350 °C. In this temperature range, the effect of pressure on the energy balance is negligible and it is recommended for the thermochemical recuperation system to use LOHC with a pressure of 1.5–2 bar. Based on the analysis, it was concluded that the temperature potential of waste heat for about 300–350 °C is enough for the investigated concept. An analysis of a mass balance showed that the decreasing in condensation temperature leads to a significant increasing in the share of condensed toluene from toluene-hydrogen mixture after a reactor. If temperature of a hydrogen-toluene mixture of 20 °C at pressure above 2 bar about 96% of toluene can be condensed after the first condenser.

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E-mail address: pashchenkodmitry@mail.ru.

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Introduction

Climate change is already affecting the Earth and everything on it. To avoid these effects becoming disastrous, the modern civilization should to keep global temperature rises no higher than 1.5° [1]. This means that within the next 10 years, global emission need to at least halve [2]. It's a big challenge for governments, societies and science. One way of tackling this challenges is by developing decarbonisation pathways. One of these pathways is developing of a sustainable hydrogen economy where hydrogen as an energy carrier is playing a core role [3,4]. Hydrogen is an important chemical precursor for many industrial applications: transport [5], chemical engineering [6], aerospace [7], etc. In a future world with renewable energy and mobility its importance as an energy source will increase considerably.

Storage and transport of hydrogen with conventional technologies have up to now been very energy intensive, inefficient, and subject to risk [8,9]. One of the ways to solve these issues is the use of liquid organic hydrogen carrier (LOHC) and thus overcomes one of the biggest challenges to the development of a sustainable hydrogen economy [10,11].

The main concept of the use of LOHC in a hydrogen economy is hydrogen generation from LOHC before use as a fuel [12]. Such way of a hydrogen generation is called an on-board hydrogen production technology [13]. Recently, the different LOHCs were proposed by engineers and scientists. The majors are the following: methylcyclohexane [14], N-ethyl carbazole [15], dibenzyltoluene [16], etc [17,18].

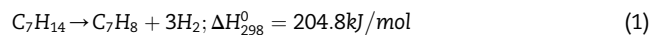
To generate hydrogen from LOHCs, the heat supply need to be organized in a catalytic reaction space where the reactions of dehydrogenation are occurring. A convenient way for heat production for the dehydrogenation reactions is the use of part of hydrogen as a fuel for heat generation [19]. A reactor for LOHC dehydrogenation is usually operating at 350 °C and 2 bar for one of the most common LOHC systems – methylcyclohexane/toluene [20]. The need to produce heat for the dehydrogenation reactions significantly reduces the efficiency of the use of LOHCs.

A moderate low-temperature potential for the dehydrogenation reaction makes it possible to use waste heat as a heat source in a reactor. In recent years, a pathway for increasing the efficiency of fuel-consuming equipment due to thermochemical waste heat recuperation has been actively developing. The main concept of thermochemical recuperation is using of waste heat for an endothermic transformation of initial fuel to produce new hydrogen-rich gas [21,22]. This new hydrogen-rich gas is utilized as fuel to generate more heat in comparison with the combustion heat of initial fuel. There are a lot of publications on thermochemical waste heat recuperation in the different fuel-consuming equipment: internal combustion engines fired with alcohols [23,24], gasoline [25], liquid oil [26,27], ammonia [28–30]; high-temperature plants fired with natural gas [31–33]; gas turbines fired with natural gas [34,35], methanol [36].

Previously, a set of optimistic studies have been published to show the efficiency of thermochemical waste heat recuperation. Poran et al. performed an experimental investigation of a direct injection internal combustion engine with

thermochemical waste heat recuperation [24]. Thermochemical recuperation was based on the steam methanol reforming process. The authors showed that an internal combustion engine with thermochemical recuperation shows a notable increase in the efficiency of 19%–30% and a significant reduction in NO_x, CO, HC and CO₂ emissions by up to 97, 91, 96 and 15%, respectively. Mardani et al. investigated synergies in renewable fuels and thermochemical exhaust heat recuperation for low carbon vehicles [29]. Thermochemical exhaust heat recuperation was implemented via reforming of butanol and ethanol with the heat supply from the gasoline direct injection engine. The authors concluded that thermochemical recuperation gives energy replacement up to 32% and 11% carbon dioxide reduction. Chakravarthy et al. performed an investigation of an internal combustion engine with thermochemical exhaust heat recuperation to determine the theoretical potentials [27]. The authors tested methanol, ethanol and isooctan as primary fuels for thermochemical recuperation. The results showed that thermochemical recuperation leads to an increase in the second law efficiency: 3% for methanol, 9% for ethanol, 11% for isooctane.

Based on the conceptual fundamentals, thermochemical recuperation (TCR) takes place when endothermic reactions occur. In this case, the enthalpy of the waste heat is converted into the chemical energy of the reaction products. For example, dehydrogenation reaction of methylcyclohexane (MCH) (1) is endothermic [37]. To obtain 1 mol of hydrogen, it is necessary to spend about 68 kJ of heat. Therefore, it can be assumed that the principles of TCR can be applied to LOHCs.



MCH dehydrogenation MCH is reversible and taking place over the catalysts. The catalytic dehydrogenation of MCH is reversible, which is affected by thermodynamics and kinetics. It is widely recognized that commonly used catalyst supports for MCH dehydrogenation include metal oxides, composite metal oxides, molecular sieves, activated carbon, carbon nanotubes and so on. For the catalyst active center, the metals such as Pt, Ir, Pd, Ru, Rh, Cr, Ni, Cu, Co, Mo, Ce, Fe, Zn and Sn are usually used as the metal core of the catalyst in the dehydrogenation of MCH [38–42].

In the convenient thermochemical waste heat recuperation systems, the main goal of reforming of initial fuel is the production of a hydrogen-rich mixture, for example, synthesis gas. Also, the whole initial fuel is converted into new hydrogen-rich fuel. After the dehydrogenation of methylcyclohexane, the reaction products consist of toluene and hydrogen. Then hydrogen can be used as a fuel and toluene can be forwarded for a hydrogenation reaction. Therefore, the hydrogen production from LOHC with the help of exhaust heat can be considered as the thermochemical recuperation process.

The publications on the use of liquid organic hydrogen carriers are mainly devoted to an investigation of LOHC compositions, chemical kinetics, catalysts preparation, etc. In the present paper, the hydrogen production from LOHC based on the thermochemical exhaust heat recuperation concept is considered. For this goal, the first law energy analysis of the TCR system based on hydrogen generation from LOHC is performed. The energy and mass balances are compiled for

the whole system and its elements. One of the main advantages of this concept is the possibility to use “green” hydrogen (hydrogen produced from renewable sources) in LOHCs, for example, in the methylcyclohexane/toluene system.

TCR concept

The main concept of thermochemical waste heat recuperation is the use of this heat for the endothermic reaction to produce hydrogen or hydrogen-rich fuel [25,31]. Fig. 1 shows a schematic diagram of the thermochemical waste heat recuperation based on the hydrogen production from liquid organic hydrogen carrier (methylcyclohexane-MCH/toluene). In this schematic diagram, the core element is reformer/reactor where the dehydrogenation reaction (1) is occurring. The reformer is operating under the following conditions: temperature of 300–350 °C, the pressure of 2–2.5 bar. At these conditions, the conversion degree of MCH to toluene and hydrogen is about unity (99.9%). One of the main advantages of the reaction of MCH dehydrogenation is no side-reactions, for example, carbon formation [43,44].

In the schematic diagram from Fig. 1, MCH from a tank at an ambient temperature is pumped by a pump from 1 bar to 2–2.5 bar. The first stage of heating is operating in the heat exchanger (H/ex) where a residual heat of a toluene-hydrogen mixture is used. After the heat exchanger, the temperature of MCH is close to evaporation temperature. To evaporate MCH, the waste heat is utilized in an evaporator. The vapor of MCH is fed to a reformer/reactor where the reaction (1) is taking place to produce a toluene-hydrogen mixture. A hot toluene-hydrogen mixture is cooling in the heat exchanger and a first condenser (Condenser1). A part of toluene is condensed and forwarded to a toluene tank. To condensate a residual part of toluene the toluene/hydrogen mixture after the first condenser is compressed up to 15–20 bar. The hot

compressed toluene/hydrogen mixture is fed to a second condenser (Condenser2). Up to 97% of toluene is condensed in the second condenser. After the second condenser, toluene is fed to the toluene tank and hydrogen is fed to a hydrogen purification system where separation is taking place to produce pure hydrogen and tail gas. Specific type of Pb-based membranes for hydrogen purification can be used in the system [45].

Thermochemical recuperation of waste heat is obtained in an evaporator and a reformer/reactor. Moreover, the schematic diagram is containing an inner recuperation system to recover the toluene/hydrogen mixture heat via heating of initial MCH. Here, it is important to state that configuration of the inner recuperation system can be considered as an important task for future investigations in this scientific field.

Thermodynamic analysis

To analyze the mass and energy flow in the thermochemical waste heat recuperation system based on the hydrogen production from LOHC, the thermodynamic analysis was performed via the software Aspen HYSYS. A computational scheme in Aspen HYSYS is presented in Fig. 2. This scheme is fully corresponding to the schematic diagram from Fig. 1. The only difference is the presence of separators after the condensers to separate gas and liquid flows.

A core element in the computational scheme is a reformer/reactor that set a Gibbs reactor. The MCH conversion degree, as well as the composition of a product gas (toluene + hydrogen mixture) and the enthalpy of the dehydrogenation process, depends on operating conditions such as temperature and pressure. An equilibrium composition of a product gas and process enthalpy was determined via Gibbs's free energy minimization method. This method is based on a fact that the chemical system is thermodynamically favorable when its total

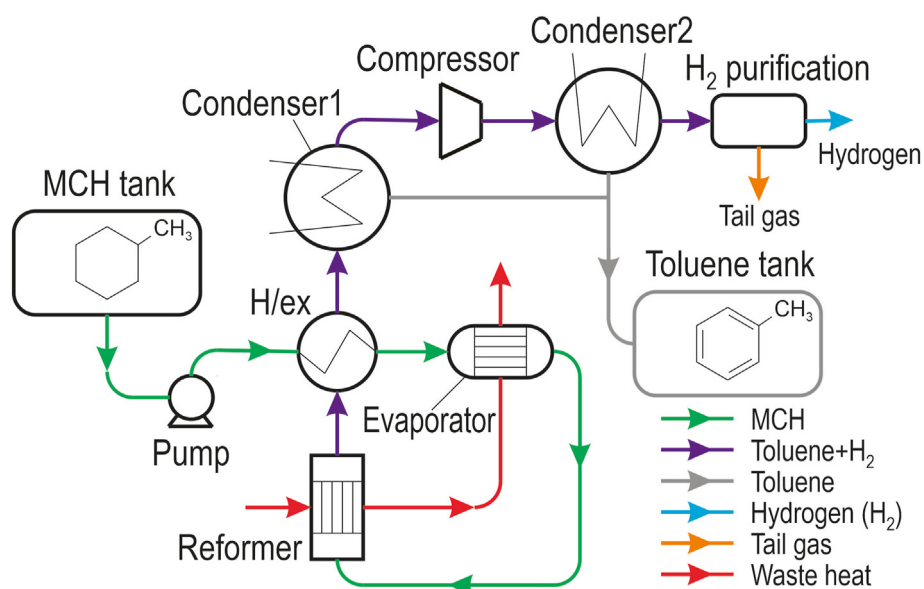


Fig. 1 – A schematic diagram of the hydrogen production system from LOHC based on the thermochemical waste heat recuperation concept.

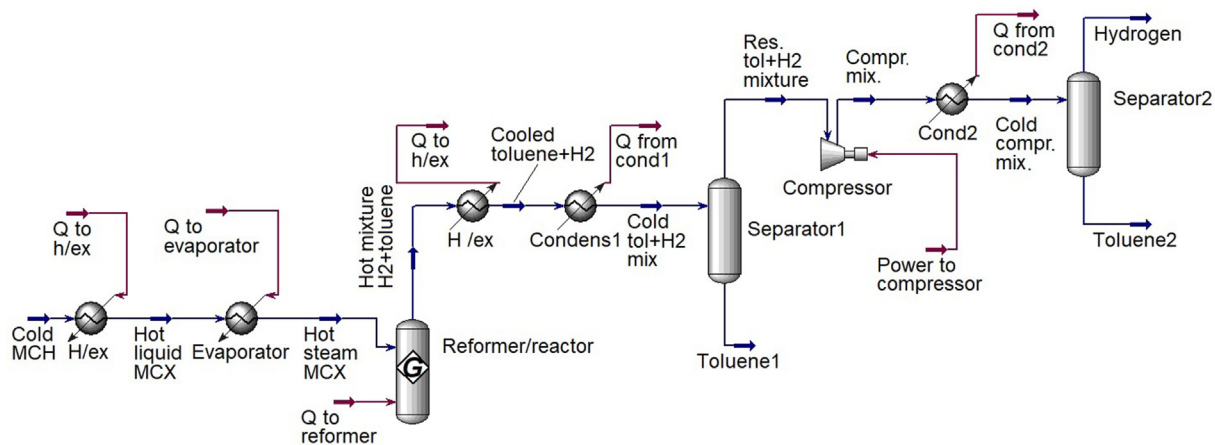


Fig. 2 – A computational scheme of the thermochemical waste heat recuperation system based on the hydrogen production from LOHC.

Gibbs free energy is minimum and its differential is equal to zero [46,47]:

$$dG^t = 0 \quad (2)$$

where G^t is total Gibbs free energy.

Total Gibbs free energy of the MCH dehydrogenation process can be calculated as a sum of the chemical potential of all components:

$$G^t = \sum n_i \mu_i \quad (3)$$

where n_i is number of i th component moles; μ_i is chemical potential of i th component.

A chemical potential of each component can be calculated based on following expression:

$$\mu_i = \Delta G_{fi}^0 + RT \ln \left(\frac{f_i}{f_i^0} \right) \quad (4)$$

where G_{fi}^0 is standard Gibbs free energy of i th component formation; R is molar gas constant; T is system temperature; f_i is fugacity of a pure i th component; f_i^0 is fugacity of a pure i th component at the standard state.

If the equations presented above will be combined, the equation for total Gibbs free energy of the chemical system can be written as the following:

$$G^t = \sum n_i \Delta G_{fi}^0 = \sum n_i RT \ln \left(\frac{f_i}{f_i^0} \right) \quad (5)$$

Obviously, the efficiency of the thermochemical waste heat recuperation process depends on the conversion degree of MCH. The equilibrium conversion degree of MCH is determined as:

$$X_{C_7H_{14}} (\%) = \frac{[C_7H_{14}]_{in} - [C_7H_{14}]_{out}}{[C_7H_{14}]_{in}} \times 100\% \quad (6)$$

where $[C_7H_{14}]_{in}$ and $[C_7H_{14}]_{out}$ is mole fraction of MCH at the reactor inlet and outlet, respectively.

Aspen HYSYS is a widely used chemical simulator of chemical and energy processes. This software is able to perform many of the core calculations of chemical

engineering, including those devoted to energy and mass balances, vapor-liquid transition, chemical equilibrium, and pressure drop. Therefore, the high-quality results are obtained from this software.

Results and discussion

Equilibrium composition and enthalpy

One of the main advantages of MCH using as LOHC is no side reactions. All mass flow MCH is converted to hydrogen and toluene with residual MCH. Fig. 3 shows the MCH conversion degree (a) and mole fraction of MCH and hydrogen (b) as a function of temperature at the different pressures. The pressures from 1 to 4 bar were tested. In the TCR based on MCH dehydrogenation, the pressures above 4 are not appropriate because an increase in pressure leads to decreasing in MCH conversion as well as decreasing in hydrogen yield. Fig. 3a shows that in the temperature range above 300 °C the MCH conversion is trending to 1. Therefore, the maximum MCH conversion is achieved at 300–350° for all investigated pressures. On the other hand, in the temperature range below 300 °C the effect of pressure is significant. While MCH conversion at 4 bar and 250 °C is 34%, MCH conversion at 1 bar for the same temperature is 87%. Fig. 3 also depicts that in the temperature range above 300 °C the pressure doesn't affect MCH conversion. Therefore, for the thermochemical waste heat recuperation systems based on MCH dehydrogenation is enough to have pressure for covering pressure losses in the schematic diagram from Fig. 1. In the practice, the pressure level in the system is no more than 2 bar. An increase in pressure will lead to an increase in work consumption for a pump.

Fig. 3b shows the mole fractions of MCH and hydrogen. Fig. 3 makes it clear that hydrogen reaches a maximum mole fraction of 0.75 faster at lower pressure in comparison to a higher pressure. The effect of pressure on the dehydrogenation process can be explained by Le Chatelier's principle because the total mole of products is higher than the total mole of initial MCH. Moreover, Fig. 3 depicts that in the

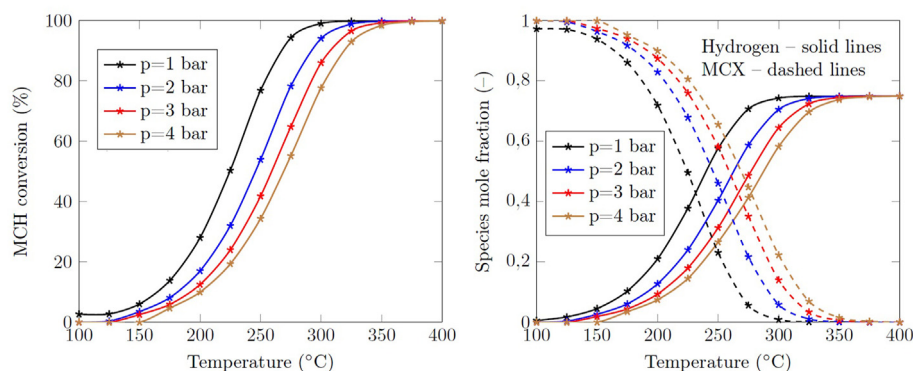


Fig. 3 – MCH conversion degree (a) and mole fraction of MCH and hydrogen (b) as a function of temperature at the different pressures.

temperature range above 300 °C the dehydrogenation products consist of toluene (0.25) and hydrogen (0.75).

As it was mentioned above, the main concept of thermochemical recuperation is a transformation of waste heat into chemical energy of hydrogen-rich fuel or hydrogen (in the case of the MCH system). The enthalpy of the endothermic process shows the waste heat that transformed into chemical energy. Fig. 4 present the dependence of the enthalpy of the MCH dehydrogenation process for each mole of hydrogen as a function of temperature at the different pressures. One of the main criteria of efficiency of organic compounds for use as LOHC is the reaction enthalpy to generate 1 mol of hydrogen. For MCH this enthalpy is 68 kJ/mole_{H₂}. The reaction enthalpy for conversion of 1 mol of MCH is about 204 kJ/mole_{MCH}.

Energy balance

To compile the heat balance of the thermochemical waste heat recuperation system based on the dehydrogenation reaction of MCH, the following initial data were accepted: mass flow rate of MCH is 10 kg/h, the inlet temperature is 20 °C, the temperature of toluene/hydrogen mixture after the condensers is 50 °C.

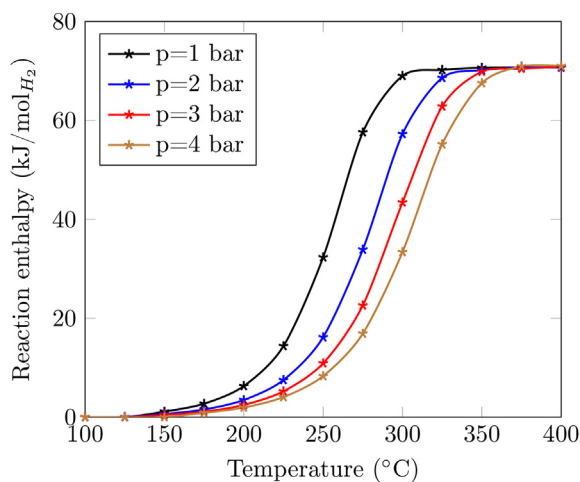


Fig. 4 – Enthalpy of MCH dehydrogenation process as a function of temperature at the different pressures.

Fig. 5 shows the Sankey diagram for the thermochemical waste heat recuperation system based on the dehydrogenation reaction of MCH at $p = 2$ bar and temperature in a reactor of 350 °C for MCH mass flow rate of 10 kg/h. The sizes of each arrow (flow) are scaled according to their value of power. In this diagram, the following values are determined:

- $\Delta H_{H/ex} = 0.433$ kW is the heat power recovered in a heat exchanger;
- $\Delta H_{H/ex} = 2.497$ kW is the heat power supplied from waste heat in an evaporator;
- $\Delta H_{H/ex} = 6.071$ kW is the heat power supplied from waste heat in a reformer-reactor;
- $\Delta H_{cond1} = 2.383$ kW, $\Delta H_{cond2} = 0.949$ kW is the heat power realized in the condenser 1 and 2, respectively;
- $W_{compressor}$ is the power consumed in a compressor to increase a pressure of toluene/hydrogen mixture up to 10 bar;
- $H_{H_2} + \Delta H_{heat-to-chem}$ is the total energy flow with hydrogen where $\Delta H_{heat-to-chem}$ is value of waste heat transformed to chemical energy.

To understand the effect of pressure on the energy balance, the heat supplied from waste heat in an evaporator and reformer was determined. Fig. 6 shows the power supplied in an evaporator (a) and total power from waste heat in an evaporator and reformer (b) as a function of temperature at various pressures. Fig. 6 makes it clear that in the temperature range above 200 °C the effect of pressure on power consumption is negligible. Moreover, the total power supplied from waste heat in an evaporator and reformer is insignificant differing due to the effect of pressure on the reaction enthalpy. Fig. 6 allows us to conclude that the effect of pressure in the thermochemical waste heat recuperation system on the energy balance is insignificant in the temperature range above 200–250 °C.

Mass balance

To compile the mass balance, the results of the thermodynamic analysis were used to understand the mass flow rates of each stream. Fig. 7 shows the mass flow rates for each stream and its temperature in the thermochemical waste heat recuperation system. The initial mass flow rate of MCH is

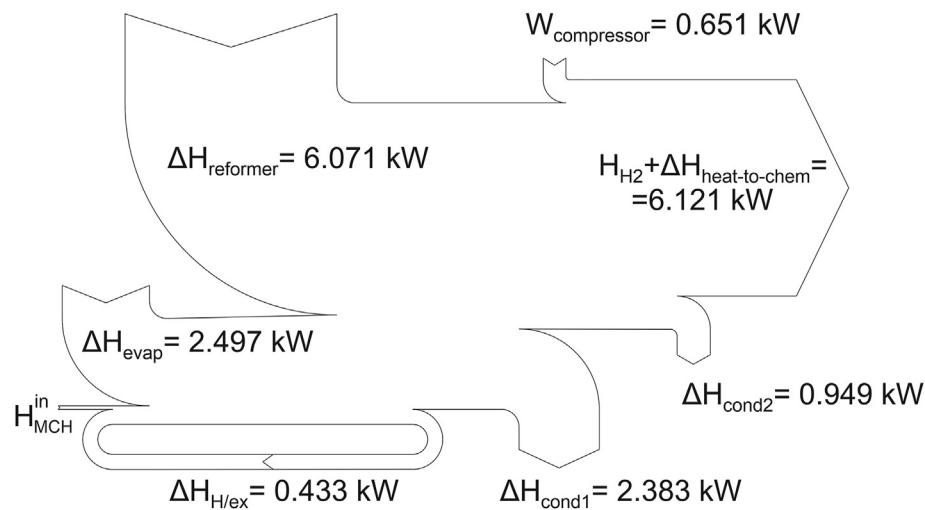


Fig. 5 – Sankey diagram for the thermochemical waste heat recuperation system based on the dehydrogenation reaction of MCH at $p = 2$ bar and temperature in a reactor of 350°C for MCH mass flow rate of 10 kg/h .

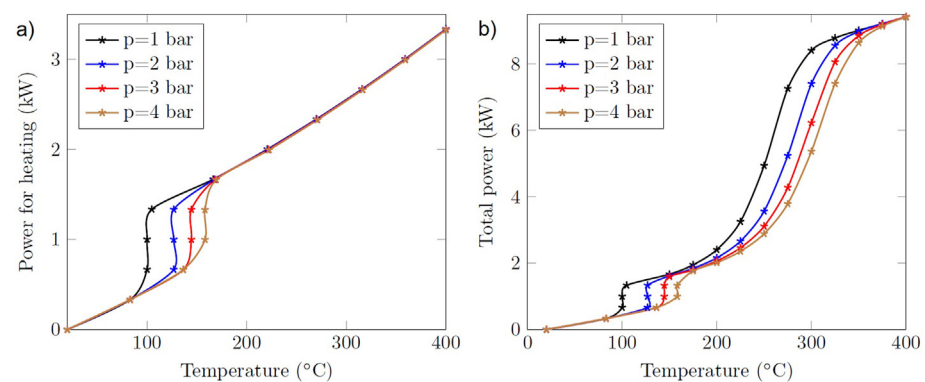


Fig. 6 – Power supplied in an evaporator (a) and total power from waste heat in an evaporator and reformer (b) as a function of temperature at various pressures.

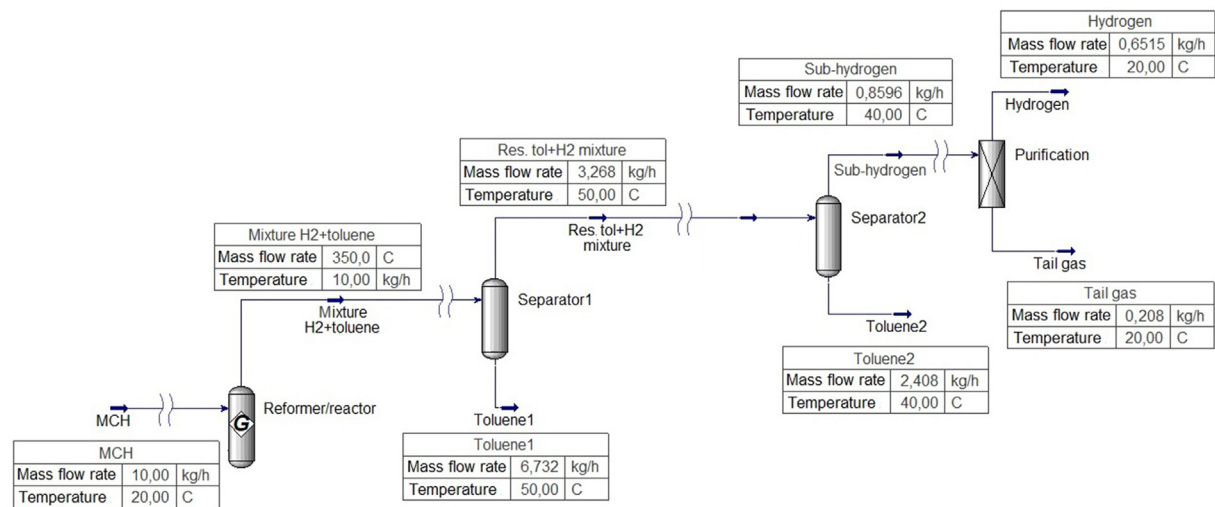


Fig. 7 – The mass flows for each stream in the system.

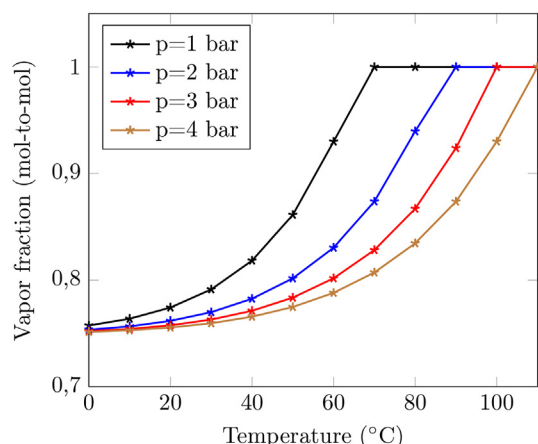


Fig. 8 – Vapor fraction in hydrogen-toluene mixture as a function of temperature and pressure.

10 kg/h. After reformer, the mixture of hydrogen and toluene with a mass flow rate of 10 kg/h is obtained. Then this mixture is cooled in the condenser1 up to 50 °C. In the Separator1 about 72% of toluene was condensed. The mixture of residual toluene and hydrogen is compressed up to 10 bar and then cooled up to 40 °C. In Separator2, about 25% of the total mass flow rate of toluene was condensed. After Separator2, sub-pure hydrogen (hydrogen with toluene vapor in a molar ratio of approximately 99:1) is obtained. In the case of using this hydrogen as a fuel, no purification is needed. This hydrogen can be used for direct combustion. However, to get high-purity hydrogen, additional purification is needed. The total mass flow rate of pure hydrogen is 0.65 kg/h that corresponding to 6.5% of the total mass of MCH. A tail gas after purification system can be used for combustion.

The hydrogen extraction system can be simplified to avoid the second condenser and compressor. The share of liquid toluene depends on the temperature in the first condenser. Fig. 8 shows the dependence of vapor fraction in the hydrogen-toluene mixture as a function of temperature and pressure. Fig. 8 makes it clear that the temperature of the mixture after the first condenser plays an important role in the share of condensed toluene. In the temperature range below 20 °C at a pressure above 2 bar about 96% of toluene can be condensed after the first condenser and separated in the first separator. Therefore, the decrease in condensation temperature is an important issue for simplification of the thermochemical waste heat recuperation system based on LOHC dehydrogenation. In the conclusion, the condensation temperature is determined by the temperature of the low-potential heat source.

Conclusion

To understand the heat and mass flows in the thermochemical waste-heat system based on an endothermic dehydrogenation reaction of liquid organic hydrogen carrier (methylcyclohexane(MCH)/toluene), the thermodynamic analysis under various operating conditions was performed

via Aspen HYSYS. It was established that in the temperature range above 300 °C the MCH conversion is trending to 1 for pressure of 1–4 bar. The maximum hydrogen yield is observed in the temperature range above 300 °C. The MCH conversion has an effect on reaction enthalpy. The maximum reaction enthalpy of 68.1 kJ/molH₂ is observed in the temperature range above 320–350°. Therefore, it can be concluded that the temperature potential of waste heat for about 300–350 °C is enough for the thermochemical recuperation. An effect of pressure on the heat balance in the pressure range of 1–4 bar is negligible in the temperature range above 300 °C. Therefore, it is recommended for the thermochemical recuperation system to use LOHC with a pressure of 1.5–2 bar. An analysis of a mass balance showed that about two-thirds of toluene can be condensed after the first condenser with cooling up to 50 °C and another one-third can be condensed in the second condenser after compressing up to 10 bar and cooling up to 40 °C. It was determined that the decreasing in condensation temperature leads to a significant increasing in the share of condensed toluene. If temperature of a hydrogen-toluene mixture of 20 °C at pressure above 2 bar about 96% of toluene can be condensed after the first condenser. The condensation temperature is determined by the temperature of the low-potential heat source. The presented analysis showed the principal possibility of use of moderate low potential heat (up to 300–350 °C) for generation high-exergy “green” hydrogen fuel.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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